

A Two-Dimensional Moisture Diffusion Continuous Model for Simulating Dry Shrinkage and Cracking of Soil

Chengzeng Yan¹; Hui Ma²; Zhicheng Tang³; and Wenhui Ke⁴

Abstract: Dry shrinkage cracking of soil is a common phenomenon in nature. The shrinking and cracking of soil have been linked to several geological disasters. To simulate dry shrinkage cracking in soil, we present a continuous 2D moisture diffusion model, which is implemented in a GPU parallel multiphysics finite-discrete element software, namely MultiFracS. In this model, the moisture distribution of the system is analyzed initially, then the shrinkage stress caused by the change of moisture is calculated. Finally, the shrinkage stress is applied to the governing equation of FDEM to calculate the mechanical fracture. By repeating the preceding procedures, the soil shrinkage cracking problem can be simulated and investigated. We first verify the correctness of the model in dealing with the moisture diffusion problem in a continuous medium. Then we study the soil dry shrinkage and cracking, and compare the simulation results with the experimental results. The simulation results are in good agreement with the analytical solution or the existing literature results, which verifies the effectiveness of the 2D moisture diffusion continuous model to simulate the dry shrinkage cracking of soil. DOI: 10.1061/(ASCE)GM.1943-5622.0002570. © 2022 American Society of Civil Engineers.

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Introduction

Dry shrinkage and cracking of soil are very common in nature. Due to the dry climate, the moisture in the soil evaporates and cracks occur on the soil surface (Goehring et al. 2015; Morris et al. 1992). The cracks caused by dry shrinkage cracking lead to significant changes in permeability and strength of the soil, thus reducing the structural stability of soil and affecting its performance (Konrad and Ayad 1997; Trabelsi et al. 2012). For example, clay liners are usually used to store hazardous and solid wastes. The liner may crack during drying so that it cannot be used as a hydraulic barrier (Omidi et al. 1996). In pavement engineering, the weakening of the soil mechanical properties caused by cracks affects the long-term performance of the road (DeCarlo and Shokri 2014; Guo et al. 2018). In addition, dry cracks cause slope instability and geological disasters such as landslides and mudslides (AghaKouchak et al. 2015; McKay et al. 1993; Omidi et al. 1996; Senior 1981; Van Baars 2005; Yesiller et al. 2000). As a result, it is critical to investigate the soil dry shrinkage cracking.

Soil is a highly complex material. Its cracking behavior is restricted by many factors such as mineral composition, boundary conditions, soil thickness, and so on (Nahlawi and Kodikara

2006; Rodríguez et al. 2007; Tang et al. 2008, 2010; Vogel et al. 2005; Yan et al. 2021a). Many scholars carried out in situ or indoor experiments to comprehensively study the phenomenon of soil shrinkage cracking. Tang et al. (2011) and Kodikara and Costa (2013) pointed out that the tensile stress increases with the loss of moisture content. When the tensile stress exceeds the tensile strength of the soil, the soil may lead to cracking. Groisman and Kaplan (1994) and Zeng et al. (2019) studied the influence of container geometry and boundary roughness on soil cracking. Lin et al. (2021) and Gui et al. (2016) discussed the influence of interfacial friction coefficient on dry shrinkage cracking of soil and found that with the friction coefficient increases, the overall morphology of the cracks is similar, and the number of cracks remains unchanged.

In addition, many scholars also study the dry shrinkage cracking of soil by using different numerical methods, which mainly include the continuum and discontinuous approaches. In the continuum approach, Hirobe (2018) built a model that divides dry cracking into the coupling of three physical processes of dry shrinkage, deformation, and fracture. Trabelsi et al. (2012) built a model that linked porosity evolution with suction and tensile strength which can capture crack initiation and propagation in a thin layer of dry clay. Vo et al. (2017) described the damage elastic cohesive fracture law of crack behavior by considering the coupling between hydraulic characteristics and mechanical characteristics. Maedo et al. (2020) proposed a model that can reproduce the separation of the soil sample from the mold based on a fully coupled fluid-solid finite element code. It is worth noting that although the approach based on continuum medium can solve the stress and strain in the continuous deformation stage with high accuracy, the continuum approaches are challenging to simulate the extension and intersection of complex fractures due to the singular stress at the crack tip and the need to track the growth of the crack.

In the discontinuous approach, Gui et al. (2016) used a mixed-mode viscous fracture model that combines the characteristics of tensile, compression, and shear of the materials for simulating soil shrinkage and cracking. Guo et al. (2018) and Tran et al. (2020) studied the drying soil shrinkage cracking by using a

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discrete element model and analyzed the influencing factors of crack morphology. Bui et al. (2015) extended the smooth particle hydrodynamics for studying the formation of cracks in the soil and the influence of soil thickness on crack mode. Gui and Zhao (2015) used the distinct lattice spring model to consider the interaction between pore pressure and soil matrix deformation and introduces soil heterogeneity parameters to consider the uneven drying during the drying process. Lin et al. (2021) studied the influence of different friction of base boundary on soil drying shrinkage cracking by changing the particle size of the fixed base particles. For the discontinuous approach, taking the block discrete element method as an example, although it can simulate crack propagation and intersection through the Voronoi element, its preprocessing is complicated and requires a large amount of calculation. For the particle discrete element method, it also requires tedious calibration work. In summary, some existing simulation methods have some defects.

To simulate the transition from continuum to discontinuity, Munjiza (Munjiza et al. 1999; Munjiza and Andrews 2000; Munjiza et al. 2004, 2015, 2020; Munjiza 2004; Munjiza et al. 2011) proposed the finite-discrete element method (FDEM) for simulating the whole process of deformation and fracture of solid materials. FDEM applies the finite element method for the solution of deformation and stress, the discrete element method to handle contact, and the cohesive element method to deal with fracture and fragmentation. In FDEM, cracks extend along the element boundary. There is no need to track the growth of cracks. Therefore, FDEM can handle the extension and intersection of arbitrarily complex cracks. At present, FDEM has been widely used in geomechanics (Knight et al. 2015; Lei et al. 2019; Lisjak et al. 2014; Rougier et al. 2014; Wang et al. 2022a, b). However, it's worth noting that at the beginning, FDEM can only handle pure mechanical fracture stimulation, but cannot consider the multiphysics effect of fluid, heat, and moisture diffusion. Therefore, Yan et al. developed a series of hydro-mechanical (Yan and Zheng 2016; Yan et al. 2016; Yan and Zheng 2017a, c; Yan and Jiao 2018; Yan et al. 2018, 2021b, 2022a, b), thermo-mechanical (Yan and Zheng 2017b, Yan et al. 2020, 2021c, d, 2022c), contact heat transfer (Yan et al. 2021e, 2022c), and hydrothermal (Yan and Jiao 2019; Yan et al. 2019, 2022d) coupling models for FDEM to model the fracturing problems driven by multiple physical fields. All these models combined with GPU parallel technology are integrated into a GPU parallel multiphysics finite-discrete element software, namely MultiFracS, by Yan et al. (2022d). Up to now, FDEM is mainly applied to rock mechanics problems but it has not yet been used to solve soil mechanics problems. In addition, FDEM lacks the corresponding moisture diffusion model and thus cannot simulate the dry shrinkage cracking of soil. Therefore, based on the FDEM framework, this paper develops a 2D moisture diffusion continuous model and is implemented in the software MultiFracS to simulate the dry shrinkage cracking of soil and quantitatively study the crack initiation, propagation, and moisture migration during the dry shrinkage cracking of soil.

Fundamentals of FDEM

In FDEM, the problem domain is discretized into the triangular element mesh. The joint element with zero thickness is inserted into the adjacent triangular element, as illustrated in Fig. 1. The deformation of the problem domain is characterized by the deformation of triangular elements. The concept of stress and strain in continuum mechanics is effectively retained. The crack initiation and propagation are simulated by the breaking of the joint elements, without the need to track the propagation of cracks. Before the joint element breaks (i.e., crack generation), the adjacent triangular

elements interact with each other through the joint element. There is no need for contact calculation for the triangular element connected by the unbroken joint element. If the joint element breaks, the discrete element method is used to calculate the contact of the triangular element near the broken joint element. In this way, FDEM can dynamically determine the contact between the adjacent triangular element and tremendously reduce the computing burden for contact detection and contact force when dealing with continuous-discontinuous problems. In addition, FDEM uses an explicit algorithm to solve the governing equation. The solution algorithm is relatively easy to be paralleled and thus can solve large-scale problems with the help of high-performance computing technologies such as GPU, OpenMP, and MPI. Therefore, based on this method, a 2D moisture diffusion continuous model is developed in the software MultiFracS to simulate the drying shrinkage cracking of soil.

2D Moisture Diffusion Continuous Model

FDEM method adopts the mesh shown in Fig. 1. Based on the mesh shown in Fig. 1, we create a mesh for the 2D moisture diffusion

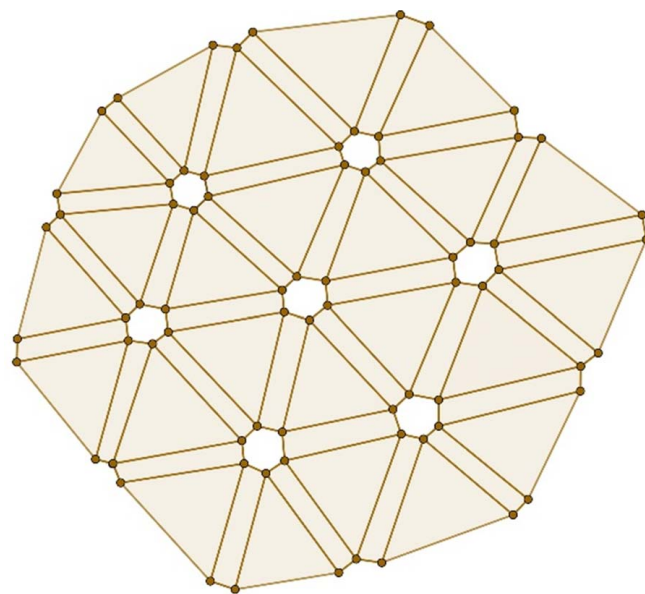


Fig. 1. Connection between the triangular element and joint element.

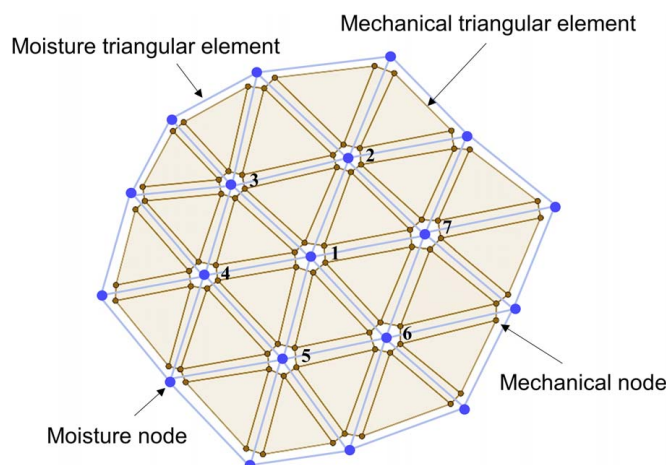


Fig. 2. Moisture field calculation schematic diagram.

continuous model, as illustrated in Fig. 2. We take the center point of the blank area surrounded by the adjacent mechanical nodes as the moisture node. The connection of the three adjacent moisture nodes constitutes the triangular element for moisture calculation. In this way, the calculation mesh of the entire problem domain is composed of moisture nodes and moisture triangular elements. Each moisture node has a mass that is one-third of the total of all linked triangle elements. The moisture content of each moisture node is used to represent the moisture content distribution of the entire problem domain. Linear interpolation may be used to calculate the moisture content of the moisture triangle element based on the moisture content of its three nodes. Taking Node 1 as an example, the triangular elements connected with Node 1 are $\Delta 123$, $\Delta 134$, $\Delta 145$, $\Delta 156$, $\Delta 167$, and $\Delta 172$. Moisture migration may occur in these triangular elements because the moisture of Node 1 may differ from that of Nodes 2, 3, 4, 5, 6, and 7. Taking the triangular element $\Delta 123$ as an example, assuming that the moisture field distribution in the triangular element is linear, the moisture gradient in the triangular element is constant and can be stated as follows:

$$\frac{\partial w}{\partial x_i} = \frac{1}{A} \int_A \frac{\partial w}{\partial x_i} dA \quad (1)$$

According to the Gauss divergence theorem, Eq. (1) can be written as follows:

$$\frac{\partial w}{\partial x_i} = \frac{1}{A} \int_s w n_i ds = \frac{1}{A} \sum_{m=1}^3 \bar{w}^m \in_{ij} \Delta x_j^m \quad (2)$$

where A = area of the triangular element; n_i = triangle element's outer normal unit vector; $\bar{w}^m$ = average moisture content at the edge m in the triangular element; Δx_j^m = coordinate difference between the two vertices of edge m ; and $\in_{ij}$ = two-dimensional displacement tensor, $\in = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Assuming that moisture migration is proportional to the moisture content gradient in an element, along the i direction, the moisture mass flow rate per unit area is expressed as follows:

$$m_i = k_{ij} \frac{\partial w}{\partial x_i} \quad (3)$$

where k = moisture conductivity coefficient tensor, which can be expressed as a function of moisture content if the relationship between them can be confirmed by experiment.

Moisture mass flow rate per unit area along the i direction can be calculated by substituting Eq. (2) into Eq. (3).

Therefore, the following formula is used to calculate the moisture mass into Node 1 through the triangular element $\Delta 123$ per unit time:

$$M_{\Delta 123 \rightarrow 1} = -\frac{m_i n_i^{(1)} L^{(1)}}{2} \quad (4)$$

where $n_i^{(1)}$ = outer normal unit vector of the edge opposite to Node 1 in the triangular element $\Delta 123$ and $L^{(1)}$ = length of the edge opposite to Node 1.

Similarly, the moisture mass into Node 1 from triangular elements $\Delta 123$, $\Delta 134$, $\Delta 145$, $\Delta 156$, $\Delta 167$, and $\Delta 172$ can be obtained as $M_{\Delta 123 \rightarrow 1}$, $M_{\Delta 134 \rightarrow 1}$, $M_{\Delta 145 \rightarrow 1}$, $M_{\Delta 156 \rightarrow 1}$, $M_{\Delta 167 \rightarrow 1}$, and $M_{\Delta 172 \rightarrow 1}$, respectively.

Therefore, the mass of water entering Node 1 in unit time is as follows:

$$M_{\text{total} \rightarrow 1} = M_{\Delta 123 \rightarrow 1} + M_{\Delta 134 \rightarrow 1} + M_{\Delta 145 \rightarrow 1} + M_{\Delta 156 \rightarrow 1} + M_{\Delta 167 \rightarrow 1} + M_{\Delta 172 \rightarrow 1} \quad (5)$$

The change in moisture content of each node can be expressed as follows:

$$\frac{\partial w}{\partial t} = \frac{M_{\text{total} \rightarrow 1}}{M_s} \quad (6)$$

where M_s = dry mass of the moisture node.

According to Eqs. (6) and (5), Node 1's moisture content can be adjusted as follows:

$$w_1^{t+\Delta t} = w_1^t + \frac{M_{\text{total} \rightarrow 1}}{M_s} \Delta t \quad (7)$$

where $w_1^{t+\Delta t}$ and w_1^t = Node 1's moisture content at the current and following time step, respectively, and Δt = time step for moisture calculation.

Node 1's moisture content at the following time step is calculated by using the preceding stages, and the moisture content in other nodes can be modified in the same way. Although the preceding equations are given for Node 1, these equations are universal; we can replace Node 1 with other nodes.

Boundary Conditions

As shown in Fig. 3, the top surface of the soil is in contact with the air, and the moisture in the soil is evaporated from the top surface at a constant rate. The evaporation rate at the soil's top surface is q , and the length of the soil boundary is L . In this way, the evaporated moisture per unit time through the top boundary of soil can be expressed as follows:

$$Q_e = qL \quad (8)$$

Then in the next time step, the moisture of the node at the top boundary of the soil is finally updated and given as follows:

$$w_1^{t+\Delta t} = w_1^{t+\Delta t} + \frac{M_{\text{total} \rightarrow 1} - Q_e}{M_s} \Delta t \quad (9)$$

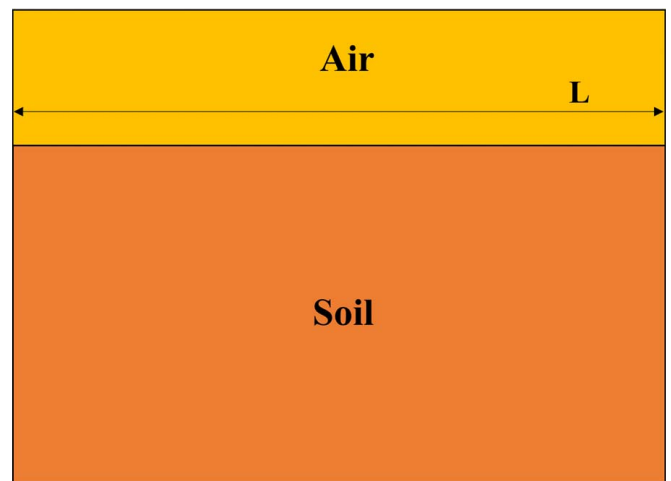


Fig. 3. Evaporation of moisture in soil through the top surface of the soil.

According to Eqs. (7) and (9), the moisture evolution in the entire region can be solved.

Moisture Diffusion-Mechanical Deformation and Fracture Coupling

Shrinkage Deformation and Stress Caused by Moisture Change

With the evaporation of moisture in the soil, the soil experiences shrinkage deformation. Due to the different moisture content changes in various parts of the soil, the moisture content in the top surface of the soil changes greatly, while the moisture content of the bottom changes relatively small. Therefore, the shrinkage deformation of top soil is larger than that of bottom soil. The inhomogeneity of shrinkage deformation induces shrinkage stress in the soil, of which the shrinkage stress at the top surface is the largest. When the shrinkage stress exceeds the tensile strength of the soil, cracks generate from the top surface of the soil and extend towards the bottom.

Assuming that soil shrinkage deformation is proportional to moisture content variation before the moisture content reaches the shrinkage limit (Hobbs et al. 2019):

$$\Delta \varepsilon_{ij} = \begin{cases} -\delta_{ij}\alpha(\omega - \omega_i), & \omega \geq \omega_s \\ -\delta_{ij}\alpha(\omega_s - \omega_i), & \omega < \omega_s \end{cases} \quad (10)$$

where α = shrinkage coefficient; ω_s = current moisture content; ω_i = shrinkage limit content; and ω_i = initial moisture (assume $\omega_i > \omega_s$).

According to the linear elastic constitutive, the stress increment caused by shrinkage deformation is calculated as follows:

$$\Delta \sigma_{ij} = \begin{cases} -\delta_{ij}3K^*\alpha(\omega - \omega_i), & \omega \geq \omega_s \\ -\delta_{ij}3K^*\alpha(\omega_s - \omega_i), & \omega < \omega_s \end{cases} \quad (11)$$

For the plane stress problem $K^* = 6KG/(3K + 4G)$ and the plane strain problem $K^* = K$, K = bulk modulus; α = shrinkage coefficient; ω = moisture content; and G = shear modulus of soil.

Changes in Elastic Modulus and Tensile Strength Caused by Moisture Change

The soil hardens when it loses moisture, from the mechanical parameters, which corresponds to the increase of the soil deformation modulus. In addition, the tensile strength of the soil increases as well. Therefore, dynamically determining the elastic modulus and tensile strength of soil according to the current moisture content in the calculation process is necessary.

The relationship between tensile strength T_u and moisture content w of Romainville clay is stated as follows:

$$T_u = 3,729e^{-(w/12)} - 25.21 \quad (12)$$

In this calculation, we use Eq. (12) to update the tensile strength of the joint element according to the current moisture content, to consider the influence of the moisture field on soil strength.

According to Cui et al. (2013), the moisture content characteristic curve of soil can be approximately expressed by the following function:

$$\lg \psi = a + b\theta_w \quad (13)$$

where ψ = suction (MPa); θ_w = volumetric moisture content (%); a and b = test parameters with values of 6.263 and -0.1052 , respectively.

Then, according to Amarasiri et al. (2011), the relationship between elastic modulus and suction can be obtained as follows:

$$E = 24\psi^{0.95}(1 - 2\nu) \quad (14)$$

where ν = Poisson's ratio of the soil, when the soil is close to saturation, then take ν as 0.42.

According to Eqs. (13) and (14), Young's modulus E can be approximately expressed as a function of moisture content, as follows:

$$E = A \cdot \exp(B \cdot w) \quad (15)$$

where $A = 3 \times 10$; $B = -0.177$; and w = moisture content (%).

Simulation of Soil Shrinkage and Cracking

Analytical Verification of Moisture Migration

We consider a rectangular plate with a size of 4 m $\times$ 2 m. The initial moisture of the plate is set to M_i^s and placed in an environment with constant moisture M^e . The upper boundary is the moisture exchange boundary, which is used to characterize the moisture change of soil in the environment of moisture M^e .

The analytical solution is given as follows:

$$M^s(x, y, t) = M^e + \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{e^{-\alpha_s r_{mn}^2 t}}{N(\beta_m)N(r_n)} \cos(\beta_m \cdot x) \cos(\gamma_n y) \cdot (M_i^s - M^e) \cdot \frac{\sin(\beta_m L_1) \sin(r_n L_2)}{\beta_m \cdot r_n} \quad (16)$$

where $\alpha_s = \frac{k_x^s}{k_y^s} \cdot \frac{1}{\rho}$; $r_{mn}^2 = \beta_m^2 + \frac{k_y^s}{k_x^s} \cdot r_n^2$; $\frac{1}{N(\beta_m)} = 2 \frac{\beta_m^2 + H_1^2}{L_1(\beta_m^2 + H_1^2) + H_1}$; $\frac{1}{N(r_n)} = 2 \frac{r_n^2 + H_2^2}{L_2(r_n^2 + H_2^2) + H_2}$; $H_1 = \frac{h}{k_x^s}$; and $H_2 = \frac{h}{k_y^s}$.

The parameters β_m and r_n are the roots of the following two transcendental equations:

$$\begin{aligned} \beta_m \tan(\beta_m L_1) - H_1 &= 0 \\ r_n \tan(r_n L_2) - H_2 &= 0 \end{aligned} \quad (17)$$

where k_x^s and k_y^s = moisture conductivity coefficients of the plate in the x and y directions, respectively; ρ_s = density of the soil; and t = time.

The calculation parameters used in the soil moisture migration model include the density of the plate $\rho_s = 1,300$ kg/m³, the moisture of the environment $M^e = 0.2$, and the initial moisture of the plate $M_i^s = 100\%$, the moisture exchange coefficient between the block and the environment is $h = 0.1$ kg/(m² · s), the moisture conductivity coefficient $k_x^s = 0.1$ kg/(m · s) and $k_y^s = 0$ kg/(m · s).

Moisture Migration at Different Times

Using the element size $L_e = 0.2$ m, the moisture distribution at different times is shown in Fig. 4(a). The numerical solution is compared with the analytical solution at different times as shown in Fig. 4(b). It can be observed that the numerical solution agrees well with the analytical solution.

Effect of Element Size on Moisture Migration

We maintain the moisture conductivity coefficient unchanged in this section, and the element size is changed to 0.2, 0.15, 0.1, and 0.05 m, respectively, as shown in Fig. 5(a). The influence of element size on the moisture migration process is discussed.

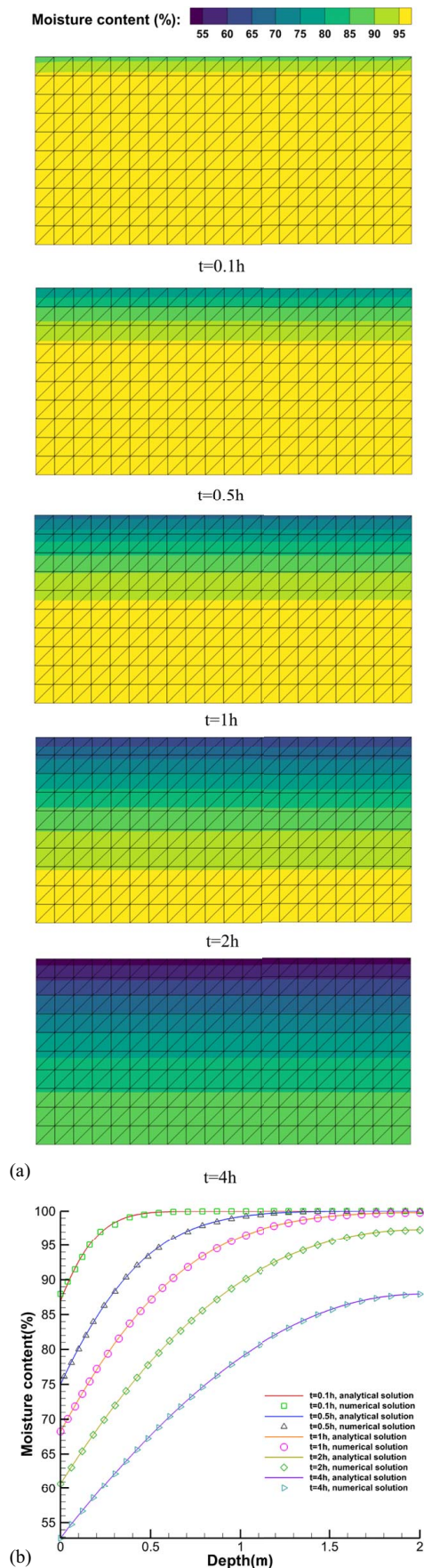


Fig. 4. (a) Moisture distribution in the soil over time; and (b) numerical and analytical solutions of soil moisture distribution along the depth direction at different times.

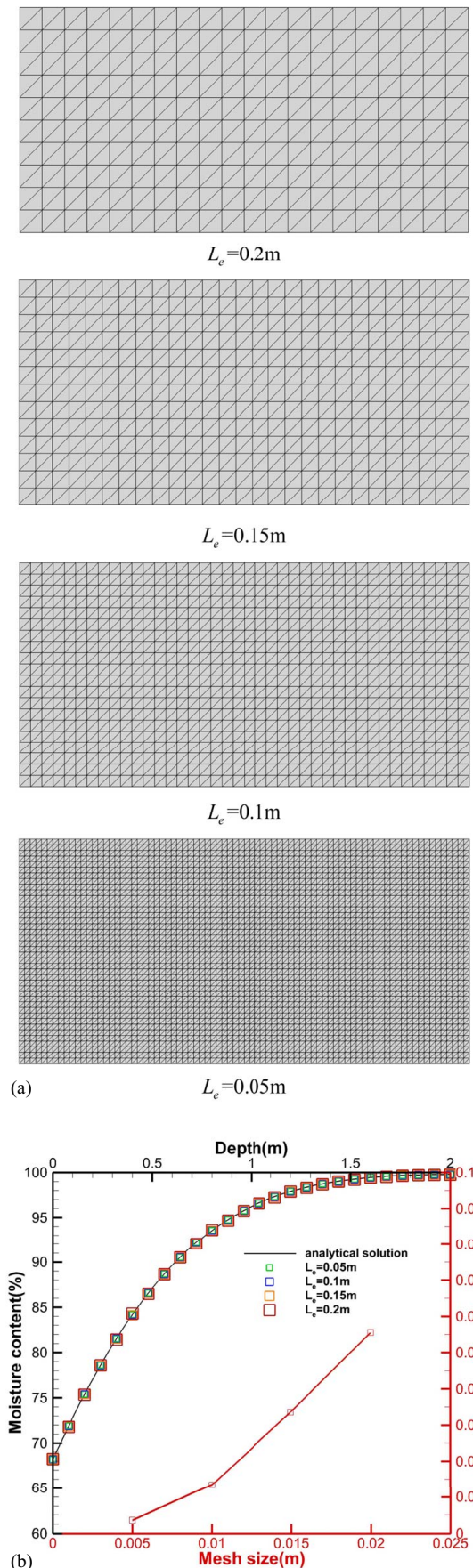


Fig. 5. (a) Four mesh with different element sizes; and (b) numerical solution for mesh with four different mesh sizes and analytical solutions at $t = 1h$ and errors in different mesh sizes.

The analytical solution of the model at $t = 1$ h is shown in Fig. 5(b). And the numerical solutions for the four meshes with varying element sizes are compatible with the analytical solution.

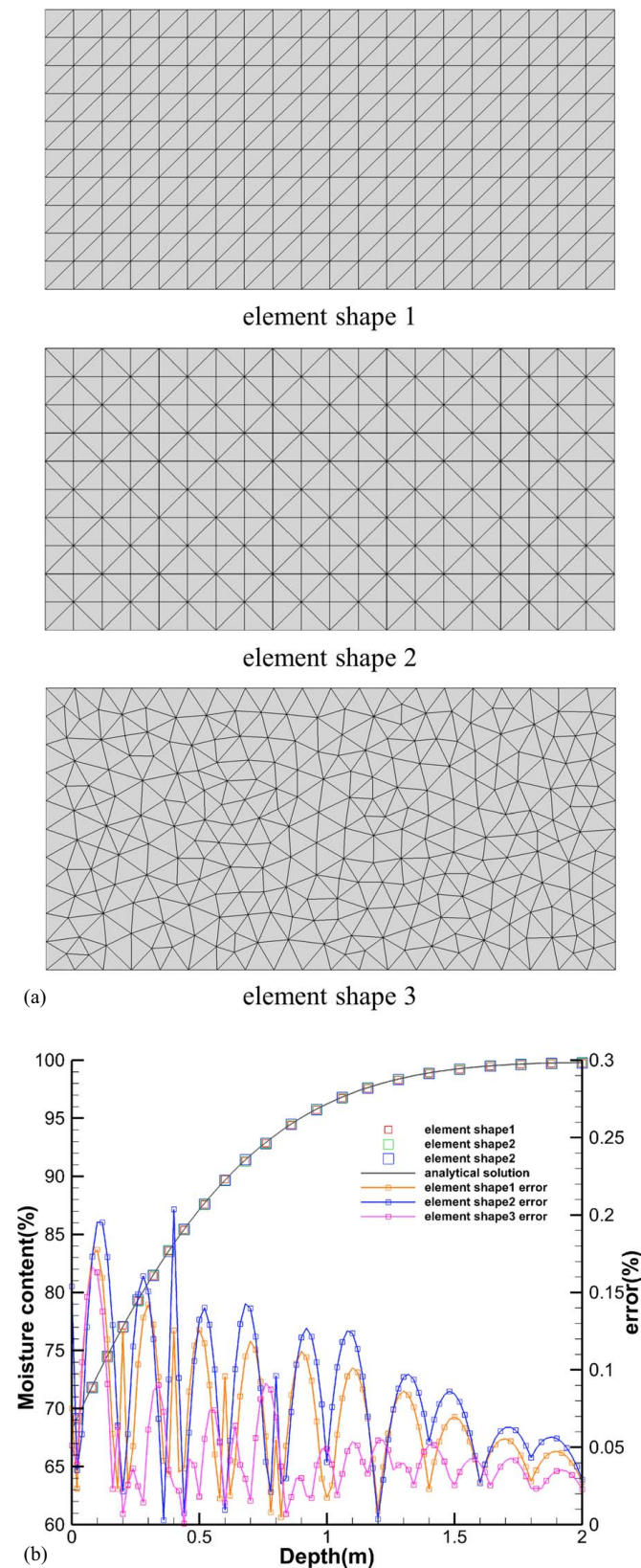


Fig. 6. (a) Three types of element shape; and (b) numerical solution for mesh with three different mesh types and analytical solutions at $t = 1$ h.

With the decrease of the element size, the error between the numerical and analytical solutions steadily diminishes.

Effect of Element Shape on Moisture Migration

We keep the moisture conductivity coefficient unchanged in this section, and the element shape is changed, as shown in Fig. 6(a). The first two meshes are relatively regular, but the third one is unstructured. It can be seen from Fig. 6(b) that the more the regularity of the element shape is, the greater the error between the numerical and analytical solutions will be. In the calculation process, the unstructured element can reduce the influence of the element shape on the calculation accuracy.

Soil Dry Shrinkage Cracking

To clarify the mechanism of dry shrinkage cracking of soil, Lin et al. (2021) selected Xiashu clay in Nanjing to conduct a series of experiments to study the dry cracking of thin clay layers. The Xiashu clay in Nanjing is poor clay with a plasticity index greater than 17 and less than 30. The initial moisture content of the sample is 70%. Sandpapers with different sand diameters are fixed at the bottom of the corresponding soil container to provide different boundary constraints, and the layer thickness of the same thickness of 10 mm is designed for each test piece.

This section first introduces the calibration of related parameters in the model. Then, the 2D moisture diffusion continuous model is employed to replicate the dry shrinkage cracking test of soil. The simulation focuses on the crack initiation and propagation and the influence of boundary constraints on crack morphology. In addition, the effects of initial moisture content, shrinkage coefficient, and evaporation rate on dry shrinkage cracking of soil are also addressed via the fracture volume in this part.

Numerical Modeling

An area with the size of 200 mm (length) $\times$ 10 mm (height) is built to represent the soil while another area with the size of 200 mm (length) $\times$ 2 mm (height) is established as a rectangular base, which is consistent with the test. In addition, the element size is set to 1 mm, the base is fixed and the upper soil is free. The interface friction coefficient between the base and the soil is set to 0.75. Both sides of the soil and the lower boundary are impervious, and the upper boundary is continuously dehydrated and evaporated. The parameters used in the simulation process are shown in Table 1.

In the drying process of the experiment, the changes of moisture content and evaporation rate with time can be divided into three stages of the drying process. In the first stage, the evaporation rate was constant, and the average moisture content of soil samples decreased from 70% to 19.5%. The evaporation rate began to decline in the second stage. The average moisture content approached

Table 1. Parameters used in dry shrinkage cracking simulation

Parameter	Soil sample	Base
Natural density (kg/m^3)	1,650	2,000
Poisson ratio	0.3	/
Elastic modulus (MPa)	0.5	50
Contact friction coefficient	0.75	/
Cohesion (kPa)	10	100
Tensile strength (kPa)	0.9	10
Moisture conductivity coefficient ($\text{kg}/(\text{m} \cdot \text{s})$)	3×10^{-4}	/
Shrinkage coefficient	0.07	/
Fracture energy I (J/m^2)	0.125	10
Fracture energy II (J/m^2)	2	10
Shrinkage limit (%)	10.5	/

the residual value in the third stage, and the evaporation rate was close to zero.

To ensure the validity and reliability of the simulation, we calculated the moisture content curve based on the evaporation rate of the experiment to keep the simulation having similar moisture content curve and the same shrinkage limit as the experiment. The variation of moisture content with time in the simulation is shown in Fig. 7. It can be seen that the evaporation rate is not constant.

The specific cracking simulation process is shown in Fig. 8(a). Soil has an initial moisture content of 70%. When the moisture content decreases to around 41.9%, cracks begin to appear at the top. When the moisture content reaches 39.8%, the number of cracks increases to four, and two of them extend obviously. When the moisture content reaches 38.4%, several tiny cracks appear at the top and one at the bottom. When the moisture content decreases to 31.1%, two cracks at the bottom can be observed obviously, and the previous bottom crack connects with the upper crack. When the moisture content reaches 24.6%, the top cracks run through the whole soil except the top cracks where the bottom cracks are located. When the moisture content reaches 12.2%, the top cracks run through the whole soil except the top cracks where the bottom cracks are located.

Fig. 8(b) shows the evolution of cracks in the process of dry shrinkage cracking. With the decrease of moisture content, the volume of cracks is also gradually increasing. When the moisture content decreases to 10.5%, the soil reaches the shrinkage limit, and the fracture volume gradually reaches stability. This is consistent with the actual process of soil shrinkage cracking.

Effect of Different Interfacial Friction Coefficients on Soil Cracking

Fig. 9(a) shows the final morphology of cracks under different interfacial friction coefficients and Fig. 9(b) shows the effect of interfacial friction coefficients on cracking when the interfacial friction coefficients are 0.75, 1, and 1.5, respectively. It can be seen from Fig. 9(a) that with the increase in friction coefficient, the overall crack morphology is similar to the end of drying, and the number of cracks remains unchanged. Lin et al. (2021) and Gui et al. (2016) also obtained similar results in their simulation. Fig. 9(b) shows that the volume of cracks generated by soil increases with the increase of interface friction coefficient. The increase of the interfacial friction coefficient means that the constraint of the boundary on the soil is stronger, and thus the crack propagation of the soil is affected. However, it is worth noting that the interfacial friction coefficient has a limited effect on soil dry shrinkage cracking.

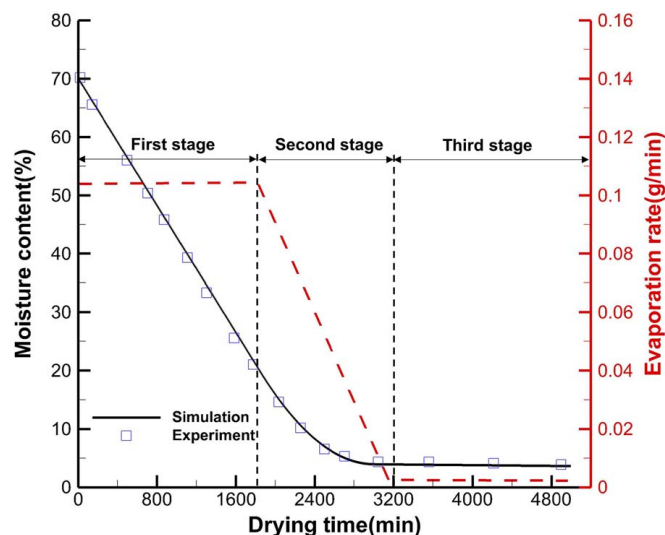


Fig. 7. Variation of moisture content with time during the simulation.

Effect of Soil Properties and Evaporation Rate on Soil Cracking

In the following study, in addition to the influencing factors of the study, other factors such as the geometric size of the soil sample, boundary conditions, and soil parameters are the same as those in the previous example. In this section, the evolution of the fracture volume over time is used to study the influence of different factors on soil cracking.

Effect of Initial Moisture Content on Soil Cracking. To study the effect of initial moisture content on dry shrinkage cracking, three sets of control tests with the initial moisture content of 75%, 70%, and 65% were set respectively. It can be seen from Figs. 10(a and b) that with the increase of the initial moisture content, both the shrinkage moisture and the fracture volume increase. The larger the difference between the corresponding initial moisture content and the shrinkage limit, the more fully the shrinkage deformation of the soil develops. But macroscopically, neither

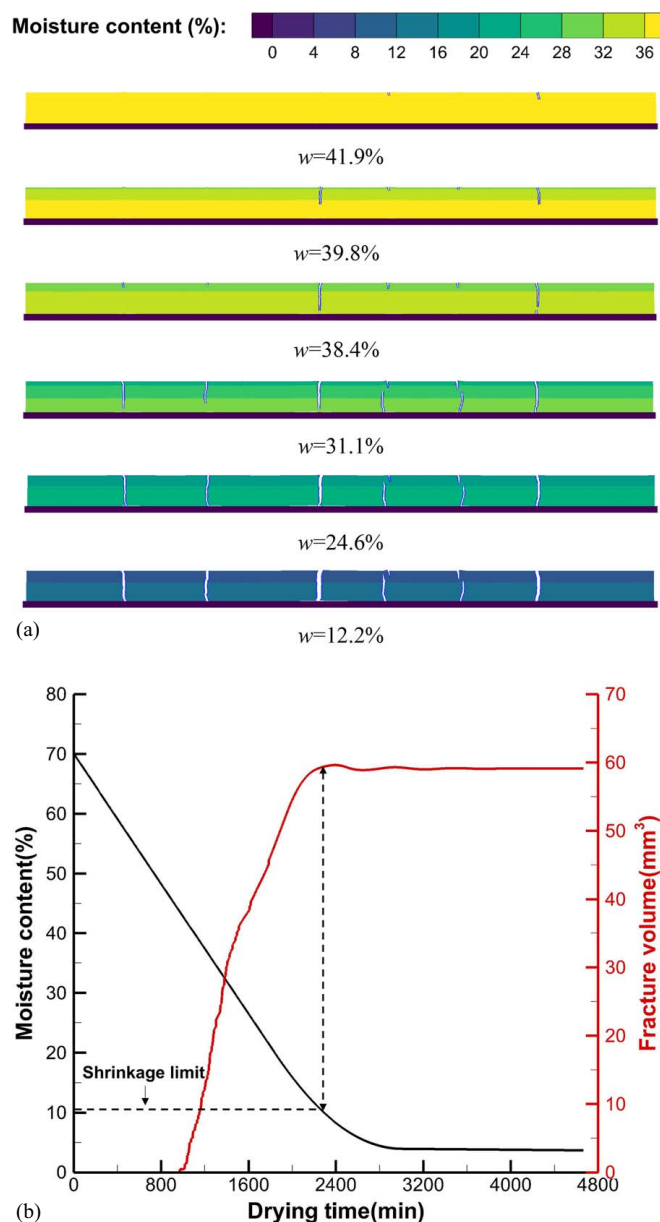


Fig. 8. (a) The evolution of moisture distribution and crack; and (b) evolution of fracture volume over time.

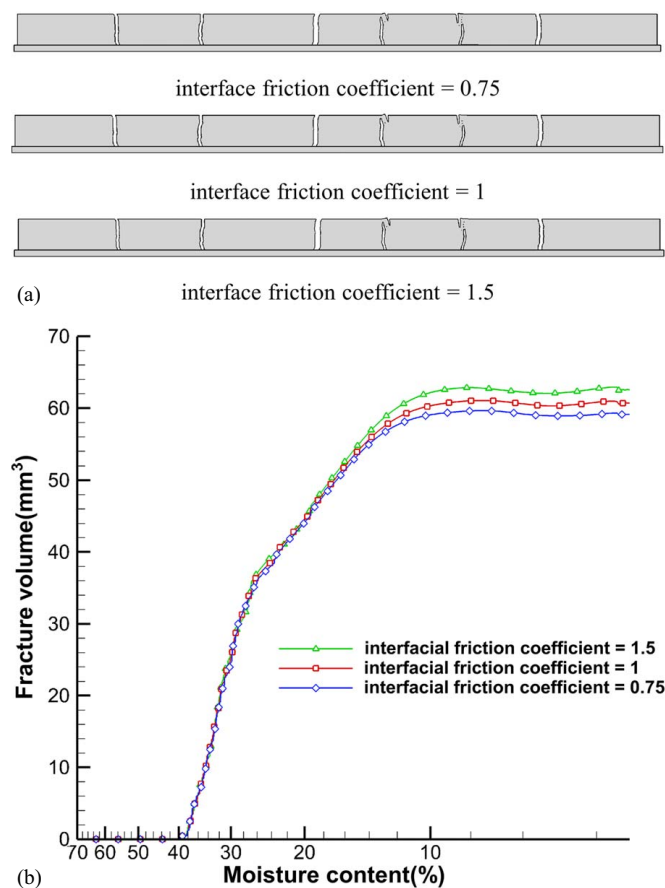


Fig. 9. Effect of different interfacial friction coefficients: (a) final morphology of cracks; and (b) evolution of fracture volume.

the number of cracks nor the depth of crack propagation has changed significantly.

Effect of Shrinkage Coefficient on Soil Cracking. Keep other parameters the same, and change the soil shrinkage coefficient, three sets of control tests with soil shrinkage coefficients of 0.06, 0.07, and 0.08 were set respectively. It can be seen from Figs. 11(a and b) With the increase of shrinkage coefficient, the initial moisture content of soil samples increases, and the fracture volume and depth increase significantly. This is because, with the increase in shrinkage coefficient, the shrinkage deformation caused by the same moisture change increase.

Effect of Evaporation Rate on Soil Cracking. Keep other parameters the same, and change the soil evaporation rate, three sets of control tests with the evaporation rates of 5.5×10^{-6} , 4.4×10^{-6} , and 3.3×10^{-6} g/s were set to study the influence of different evaporation rates on the soil cracking process. The drying moisture content of soil changes with time is shown in Fig. 12(a).

As shown in Fig. 12(b), with the increase in evaporation rate, the crack initiation time of the soil sample becomes earlier and the fracture volume increases significantly. This is because with the acceleration of evaporation rate, the greater the change in soil moisture at the same time, the greater the shrinkage deformation and shrinkage stress produced, to reach the tensile strength of the soil earlier and generate cracks earlier. In addition, with the increase in the evaporation rate, the fracture volume also increases.

Effect of Soil Thickness on Soil Cracking

Keep the other parameters the same, change the thickness of the soil, and set the soil thickness to 5, 10, and 15 mm for control

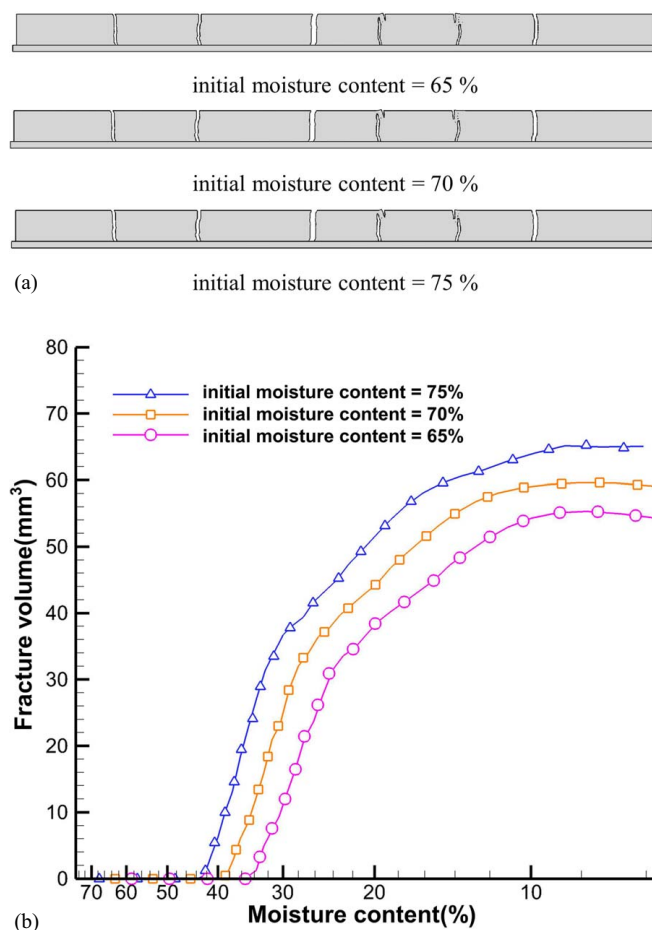


Fig. 10. Effect of different initial moisture contents: (a) final crack morphology; and (b) evolution of fracture volume.

tests. Similarly, the evaporation rate was modified in the simulation to ensure that the residual moisture content reached in the evaporation process is consistent, as shown in Fig. 13(a). The final crack patterns of different thicknesses are shown in Fig. 13(b).

It is clear that the smaller the soil thickness is, the more cracks are formed. This is because the evaporation rate increases with the decrease of soil thickness, and the shrinkage deformation caused by the same moisture change also increase. Therefore, the soil is more prone to cracking. It is worth noting that when the thickness of the soil is 5 mm, some cracks at the bottom and the upper cracks run through, so the number of cracks observed is less than the actual. It is in line with the law that the thicker the soil is, the less the number of cracks is (Zeng et al. 2019).

Free Shrinkage of Soil

The 2D moisture diffusion continuous model is utilized in this part to replicate the free shrinkage test and compare it with the experiment conducted by Tran et al. (2020). Furthermore, the effects of initial moisture content, shrinkage coefficient, and evaporation rate on soil volume during free shrinkage are considered independently. The soil volume is defined as the sum of the volume of each element to consider the change law of soil quantitatively.

Numerical Modeling

The model size is 251 mm (length) $\times$ 12.4 mm (height), and the element size is 0.8–1.2 mm. Both sides of the soil and the lower boundary are impervious, and the upper boundary is continuously

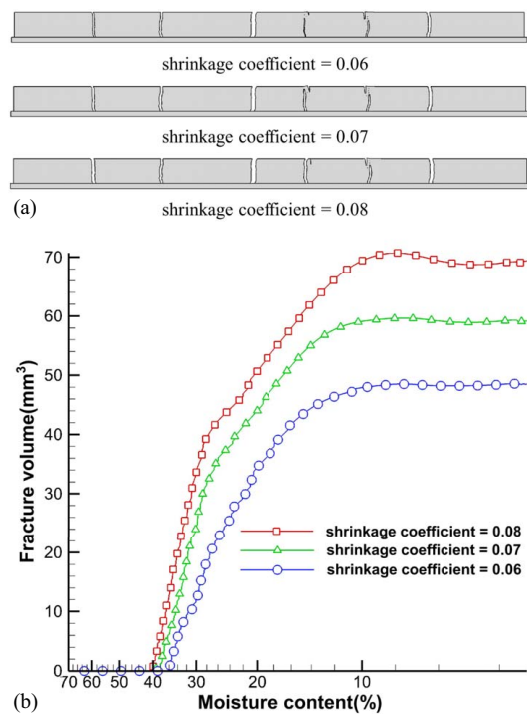


Fig. 11. Effect of shrinkage coefficient: (a) final crack morphology; and (b) evolution of fracture volume.

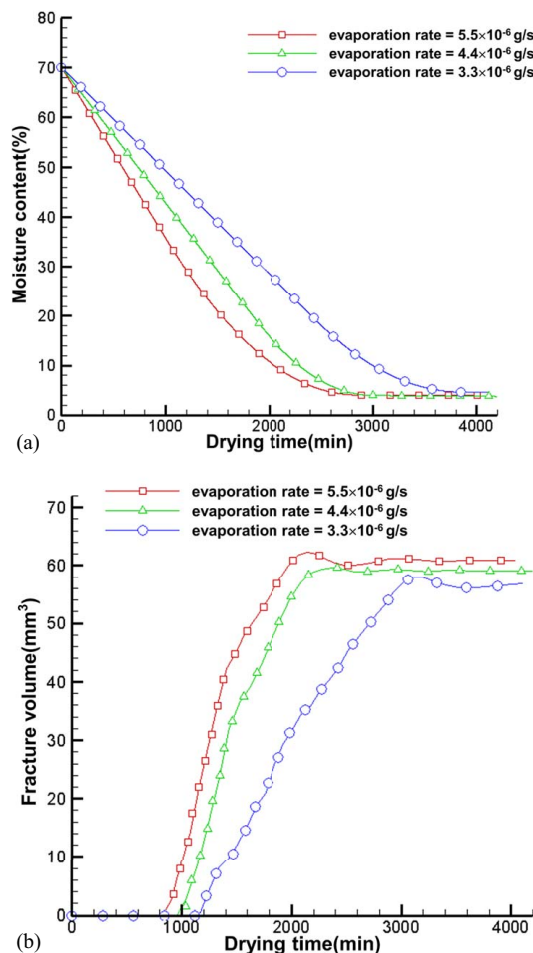


Fig. 12. Effect of evaporation rate: (a) moisture content curves at different evaporation rates; and (b) evolution of fracture volume.

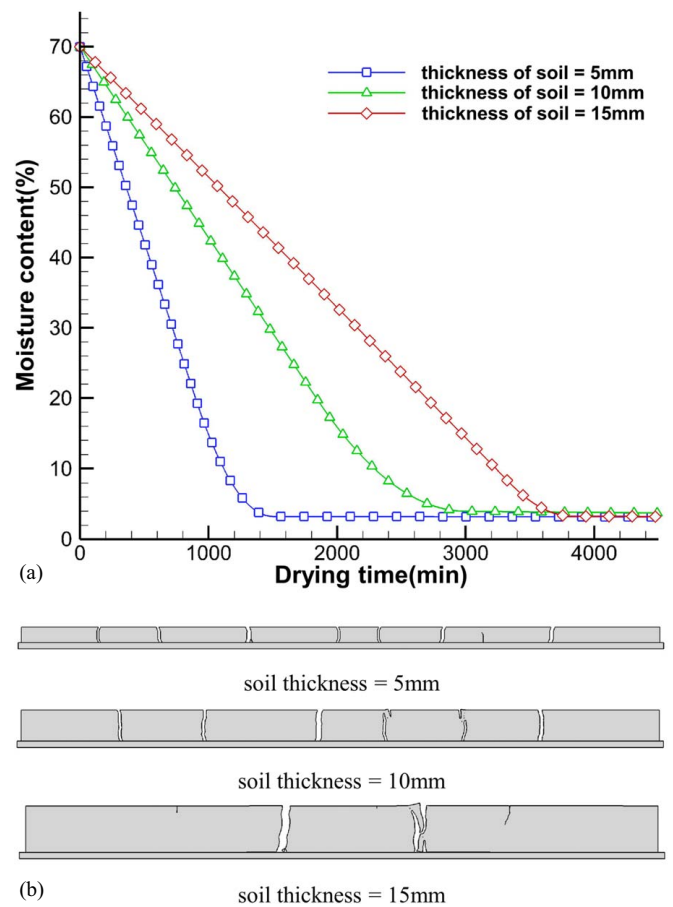


Fig. 13. Effect of soil thickness: (a) moisture content curves at different soil thicknesses; and (b) final crack morphology.

Table 2. Parameters used in free shrinkage simulation

Parameters	Soil
Natural density (kg/m^3)	1,650
Poisson ratio	0.3
Elastic modulus (MPa)	0.5
Tensile strength (kPa)	5
Moisture conductivity coefficient ($\text{kg}/(\text{m} \cdot \text{s})$)	3×10^{-5}
Shrinkage coefficient	0.14
Fracture energy I (J/m^2)	2
Fracture energy II (J/m^2)	2
Shrinkage limit (%)	29.4

evaporated. The parameters used in the simulation process are shown in Table 2.

The variation of moisture content with time in the simulation is shown in Fig. 14. The specific simulation process is shown in Fig. 15(a). It can be seen that the sample is in the initial stage of all-around shrinkage, and horizontal deformation dominates. When the moisture content decreases to about 43%, obvious vertical warping occurred on both sides of the soil sample. When the moisture content decreases to about 30%, the soil further warps and the vertical deformation is small, and the main deformation of soil is horizontal deformation. When the moisture content decreases to 11.4%, the volume does not change. This is because the soil reaches the shrinkage limit and the soil deformation reaches stability. To reflect this feature, the volume reduction of the model is counted as shown in Fig. 15(b).

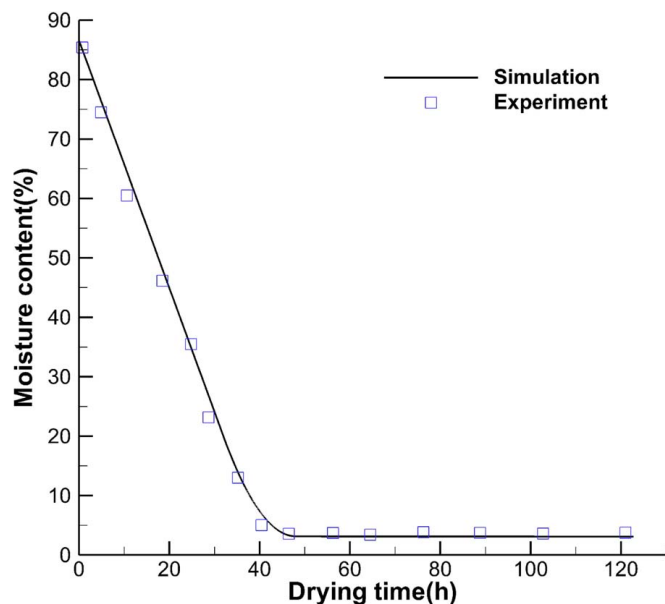


Fig. 14. Changes of moisture content with time during free shrinkage.

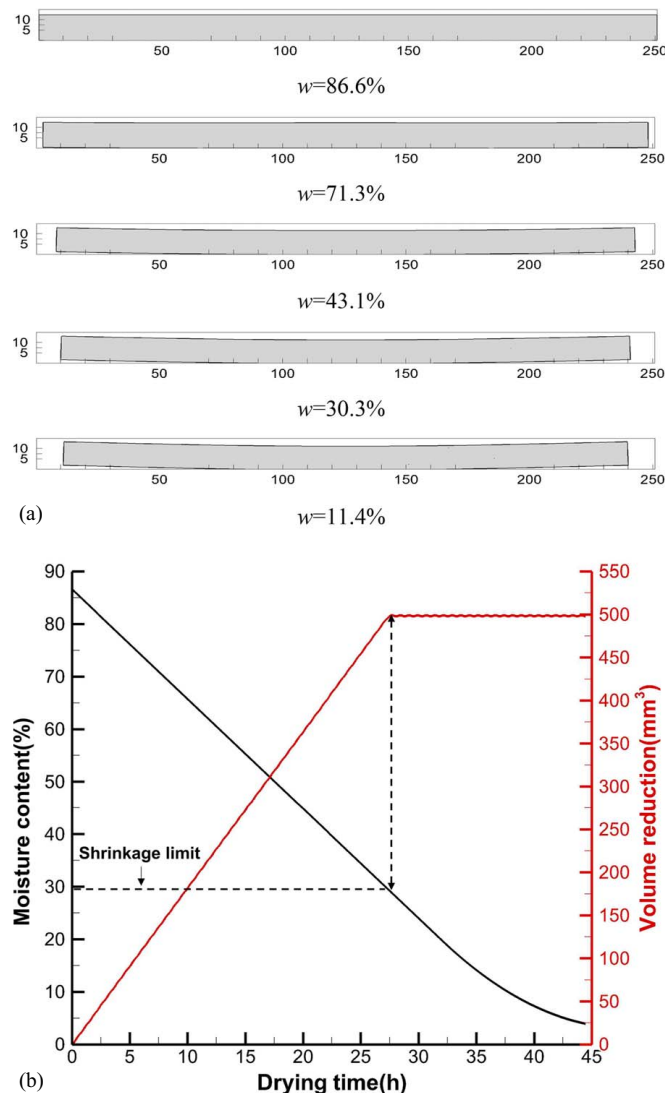


Fig. 15. (a) Simulation of free shrinkage process; and (b) soil volume reduction with time.

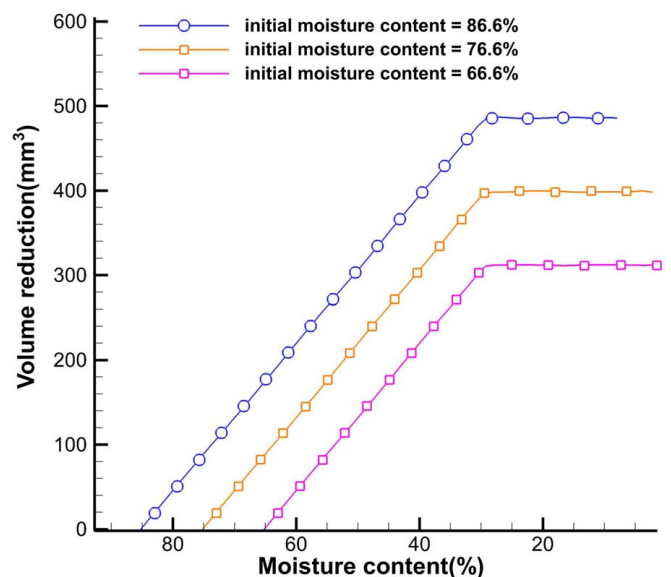


Fig. 16. Volume reduction with moisture content under different initial moisture content.

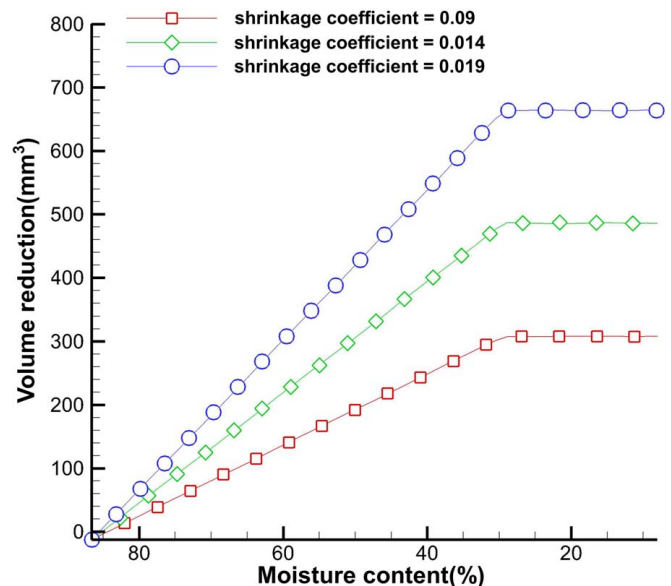


Fig. 17. Volume reduction with moisture content under different shrinkage coefficients.

Effects of Soil Properties on Free Shrinkage

In this section, the geometric dimensions, boundary conditions, and soil parameters of the soil sample are the same as in the previous example. The influence of different factors on free shrinkage is discussed through the volume reduction with the change of moisture content.

Effect of Initial Moisture Content on Free Shrinkage. Three groups of control tests with initial moisture contents of 86.6%, 76.6%, and 66.6% were set respectively. The volume reduction with moisture content under different initial moisture content is shown in Fig. 16. It can be seen from Fig. 16 that with the increase of initial moisture content, the volume reduction of the soil sample increases. This is because the greater the initial moisture content, the greater the change in the soil moisture content and the greater the shrinkage and deformation of the soil when the water loss of the soil reaches the shrinkage limit. Therefore, the greater the

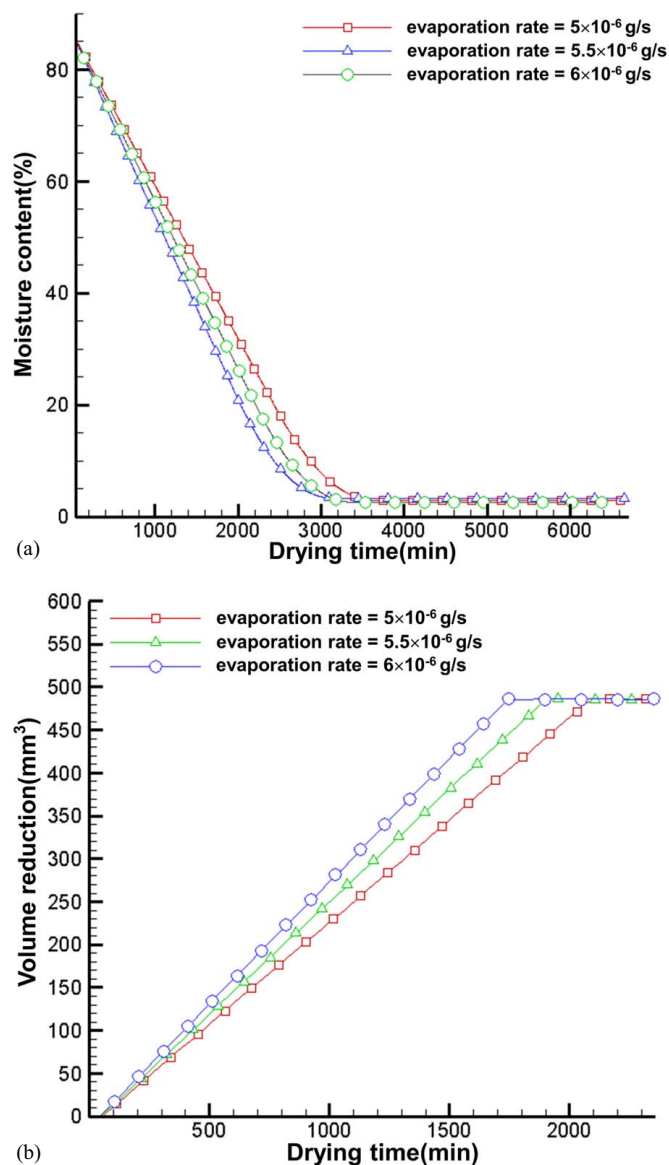


Fig. 18. Effect of evaporation rate: (a) moisture content curves at different evaporation rates; and (b) volume reduction.

initial moisture content, the greater the volume change caused by soil shrinkage.

Effect of Shrinkage Coefficient on Free Shrinkage. Set up three groups of controlled experiments with shrinkage coefficients of 0.09, 0.14, and 0.19. It can be seen from Fig. 17 that with the increase of shrinkage coefficient, the volume reduction of soil increases. This is because the shrinkage deformation caused by the same moisture change increase with the increase of shrinkage coefficient, and the volume reduction of soil also increase.

Effect of Evaporation Rate on Free Shrinkage. The evaporation rate is set as 5×10^{-6} , 5.5×10^{-6} , and 6×10^{-6} g/s for three sets of control tests to consider the influence of different evaporation rates on the free shrinkage process of the soil. Other parameters keep unchanged. The moisture content of soil changes over time as shown in Fig. 18(a). Similar to soil cracking, the evaporation rate is modified in the simulation to ensure that residual moisture content is the same for the three tests.

The volume reduction with moisture content at different evaporation rates is shown in Fig. 18(b). With the increase in evaporation rate, the volume reduction rate of soil samples also increases, but

the total volume reduction does not change. This is because with the increase in evaporation rate, the greater the moisture change per unit time, the larger shrinkage deformation produced per unit time. However, due to the same initial moisture content and shrinkage moisture content, the final soil volume reduction is the same.

Conclusion

A 2D moisture diffusion continuous model is developed in this study. The model is combined with FDEM mechanical fracture computation to create a linked model moisture diffusion-deformation-fracture model, which is then implemented in an in-house GPU parallel multiphysics finite-discrete element software, MultiFracS, to simulate soil dry shrinkage cracking. The moisture migration in the continuous medium is studied to verify the correctness of the moisture diffusion continuous model. The simulation results of indoor soil dry shrinkage cracking and free shrinkage test are given and compared with the experimental results. And the numerical results are also in good agreement with the experimental results, indicating that the coupled moisture diffusion-deformation-fracture model can predict the soil's large deformation and the crack initiation and propagation during soil dry shrinkage. Therefore, the 2D moisture diffusion continuous model and the coupled moisture diffusion-deformation-fracture model can provide a powerful tool for solving the dry shrinkage cracking problem of soil.

Data Availability Statement

All data, models, and code generated or used during the study appear in the published article.

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